

Here, 'a' stands for the discrete signals. To see the number of sam-

ples in variable 'a', we write length command 'length(a)' in the command window as shown in Fig. 9.

Considering that the handel.wav file has a sample rate of 44,100, which means that one second of audio has 44,100 samples, we divide the total number of samples by 44,100 and get the duration of audio as 1.6579 seconds.

To plot the signal, type the command given below (its output is shown in Fig. 10):

```
plot(a);
```

X-axis denotes the number of samples (0 to 80,000), and Y-axis denotes the amplitude.

Now, let us see how to manipulate the audio signal.

a1 = a; % stores the value into new variable (a1) so that we can manipulate the previous audio signal stored in variable 'a'. Let us make the amplitude zero of samples from 20,000 to 40,000 by writing the syntax as given below (its output is shown in Fig. 11):

```
a(20000:40000) = 0;
```

```
plot(a);
```

Now, play the original audio signal which is stored in variable 'a1' and play the modified audio signal in 'a' using following commands:

```
sound(a1);
```

```
sound(a);
```

While listening to signal 'a', you will find that the sound will stop in between and then start again. This shows that the samples were zero in between.

Next, we plot both the signals, the original audio signal and modified signal, to see the difference in them using the following commands (its output as shown in Fig. 12).

```
plot(a1);
hold on
plot(a, 'r');
```

Here, 'red' represents the

modified signal and 'blue' the original signal.

Voice recording, playback and plotting. To record your voice, use the following commands:

```
audio = audiorecorder(Fs, nBits,
nChannels);
```

Here, Fs is the sampling rate in Hz. Valid values for Fs depend on the specific audio hardware installed. Typical values supported by most sound cards are 8000, 11,025, 22,050 and 44,100Hz—default value is 8000Hz.

Nbits is bits per sample. Valid values for nBits depend on the operating system. For Windows, it is 8, 16, 24—default is 8.

nChannels is the number of channels: 1 (mono) or 2 (stereo)—default is 1.

For example,

```
audio = audiorecorder(8000, 8, 1);
record(audio, 5); % 5 seconds for
recording.
test = getaudiodata(audio); % Store
data in double-precision array.
sound(test);
plot(test);
```

As shown in Fig. 13, the plot is of recorded audio signal at the sampling rate of 8000Hz. Now, we will change the sampling rate to 80Hz, record and plot audio signal as shown in Fig. 14.

```
audio2 = audiorecorder(80, 8, 1);
record(audio2, 5); % 5 seconds for
recording.
test2 = getaudiodata(audio2); % Store
data in double-precision array.
sound(test2);
plot(test2);
```

Next, we plot both the audio signals of different sampling rates (8000Hz and 80Hz) on the same screen. The commands for the same are given below:

```
plot(test);
hold on
plot(test2, 'r');
```

The output plot is shown in Fig. 15. ●

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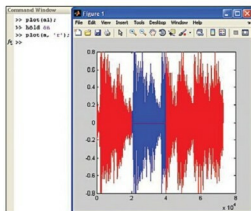


Fig. 12: The plot of original and modified audio signals

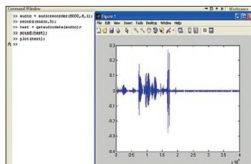


Fig. 13: Audio signal with sampling rate of 8000Hz

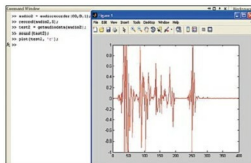


Fig. 14: Audio signal with sampling rate of 80Hz

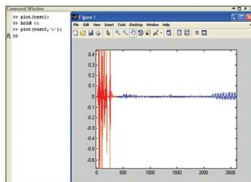


Fig. 15: Difference between 8000Hz and 80Hz sampling rate